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Interactive figures, code and this poster
kaushikmadapati7.github.io/synthetic-dce-mama

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When Regression Beats Generation: Virtual Contrast-Enhanced Breast MRI and What Its Metrics Reward

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The benchmark's eight metrics span only three to four independent axes, and FRD rewards under-enhancement, while a residual model beats matched GAN and diffusion baselines on every fidelity and tumour metric.

Problem

- DCE-MRI needs an intravenous gadolinium agent; retention is documented after repeated doses.
- Virtual contrast enhancement predicts post-contrast from a non-contrast scan.
- MAMA-SYNTH scores it with **eight metrics in four groups**.

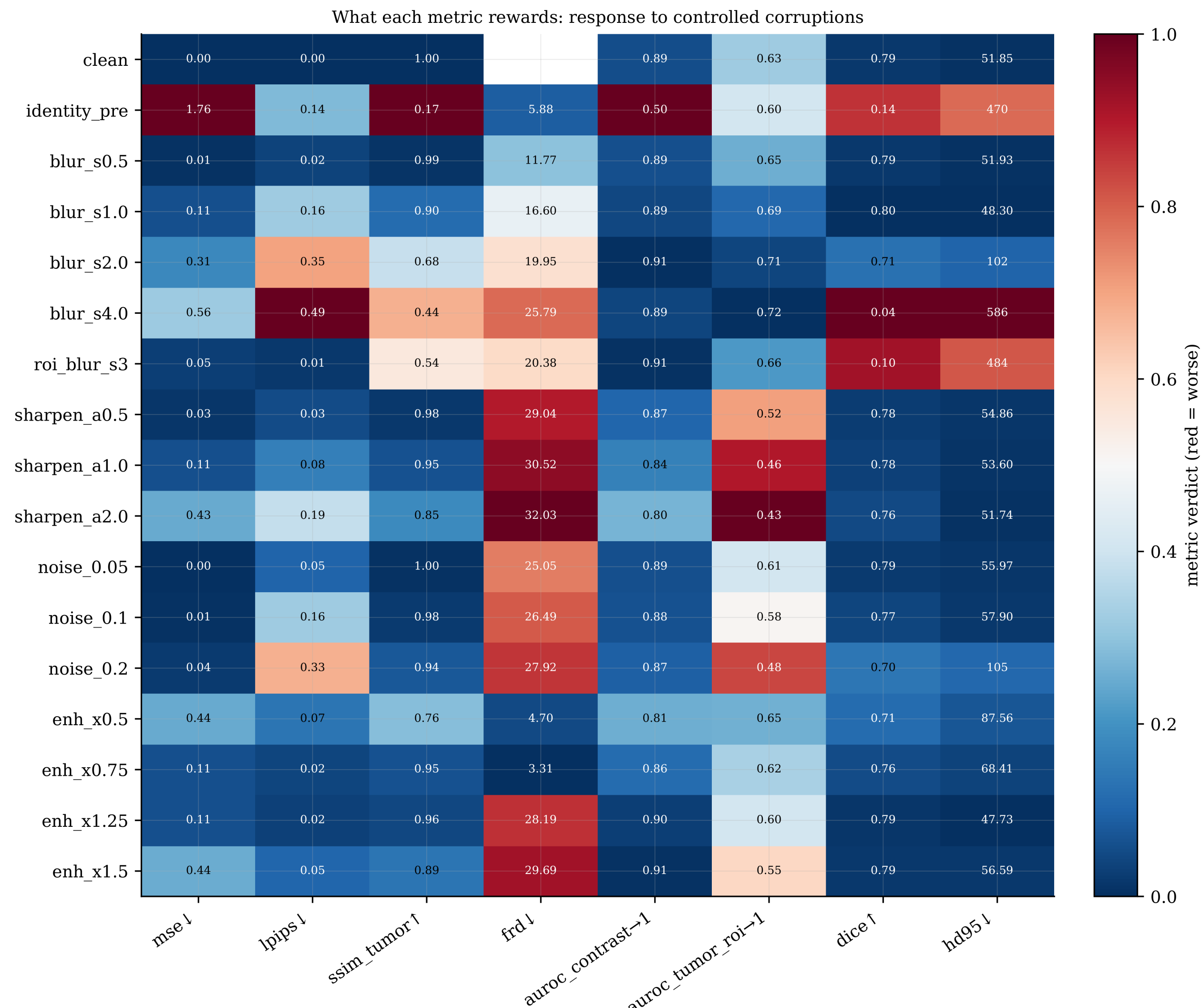
Data & protocol

- **1,506 slices** from DUKE, ISPY1, ISPY2 and NACT.
- **Stratified split** plus **leave-one-collection-out** for generalisation.

Takeaways

- **FRD rewards under-enhancement**; read it alongside the ROI metrics.
- **Eight metrics, three to four axes**: reporting all eight is not four kinds of quality.
- **Regression won at a matched budget**.
- **Select on held-out collections**; in-distribution scores are optimistic.

Main finding: what the metrics reward



3.3 vs 28.2

FRD at 0.75× vs 1.25× enhancement: under-enhancing is nearly free, over-enhancing is punished like noise

5.9

FRD of simply copying the input, worse than under-enhancing

$\rho = 0.72-0.87$

MSE, SSIM, DICE and HD95 move as one axis: eight columns, three to four independent axes

Fig. 4 Ground-truth images corrupted in controlled ways and re-scored by the official pipeline; red = judged worse. Single pass, 150 images.

Method

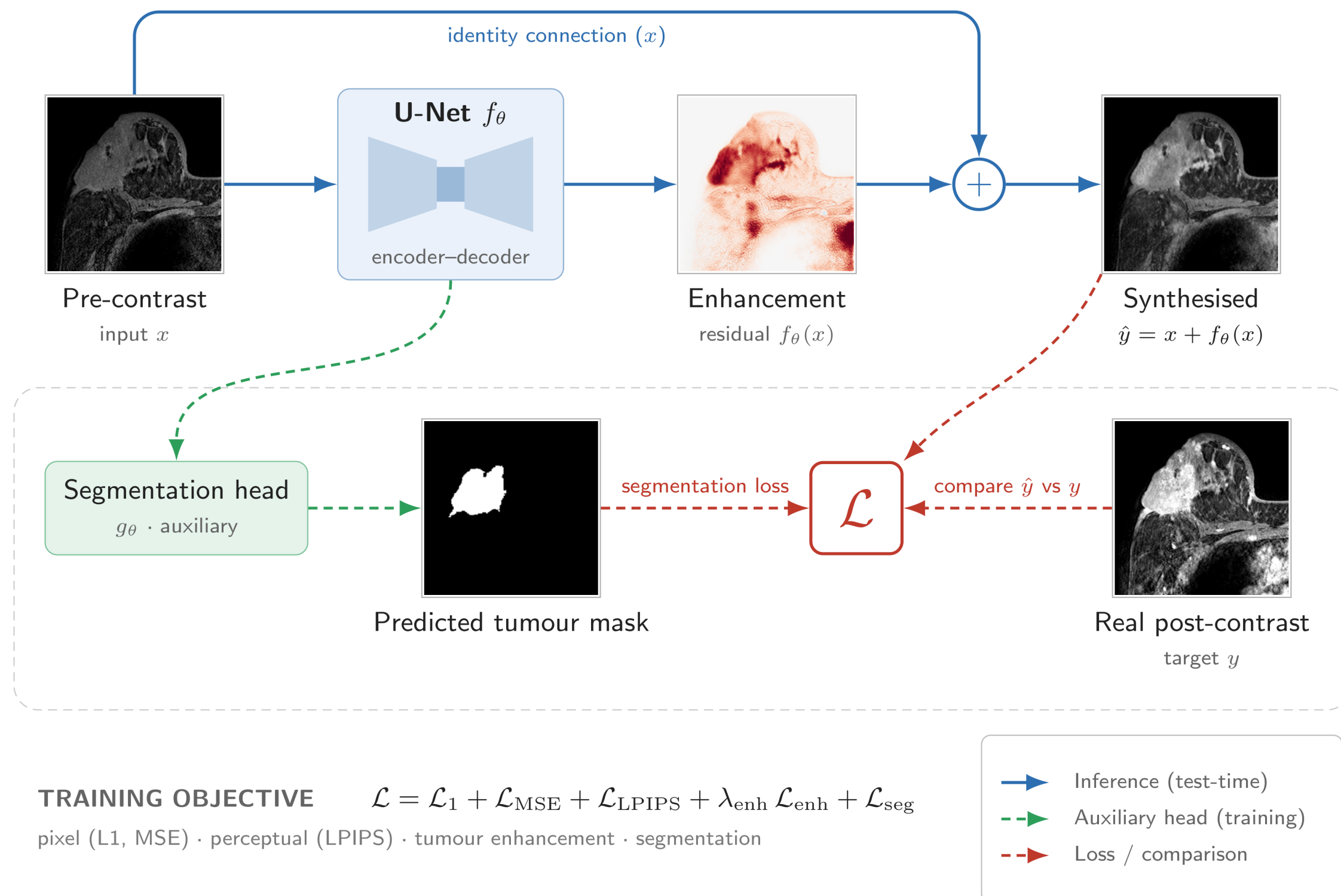


Fig. 1 Residual U-Net with an identity connection; the auxiliary segmentation head supervises training only.

Cross-domain gap

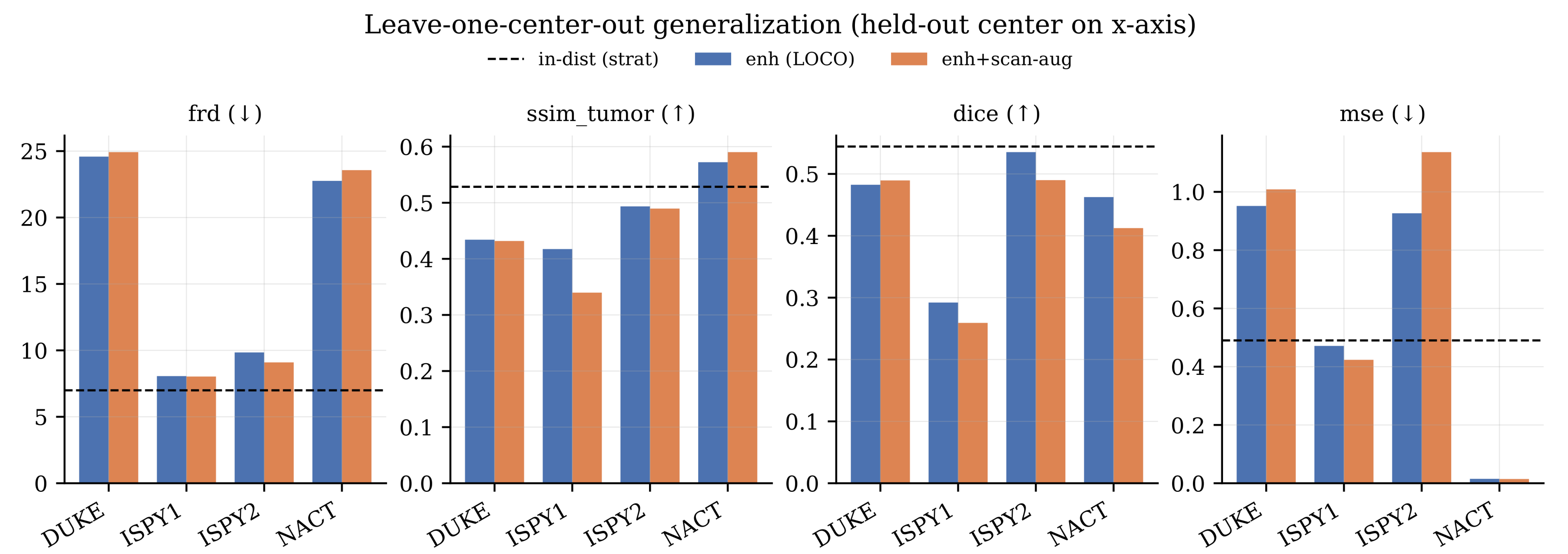


Fig. 3 Leave-one-collection-out, without (blue) and with (orange) scanner-style augmentation; dashed = in-distribution. Single runs.

FRD 7.0 → 9-25 on held-out collections, worst on the single-institution DUKE and NACT.

Augmentation does not close the gap; tumour DICE falls to 0.29 on ISPY1.

Model comparison

Method	MSE ↓	LPIPS ↓	SSIM ↑	FRD ↓	AUC-C	AUC-T	DICE ↑	HD95 ↓
Residual U-Net	0.486	0.086	0.499	7.90	0.808	0.656	0.487	120.7
Multi-task U-Net	0.476	0.087	0.534	8.79	0.761	0.677	0.555	92.5
Conditional GAN	0.894	0.192	0.199	13.68	0.830	0.610	0.127	295.1
LDM (flow)	0.969	0.159	0.246	8.62	0.888	0.635	0.210	193.5
Ours (multi-task + enh)	0.503	0.091	0.527	6.55	0.859	0.668	0.551	83.0
Ours (+ positive)	0.506	0.092	0.529	5.79	0.822	0.667	0.545	98.6

Table 1 Stratified split. Best per column in bold; our rows shaded. One configuration per generative family.

0.48 vs 0.9-1.0

MSE, regression vs generative

0.55 vs ≤ 0.21

tumour DICE, regression vs generative

5.79

best FRD, positive constraint

Qualitative results

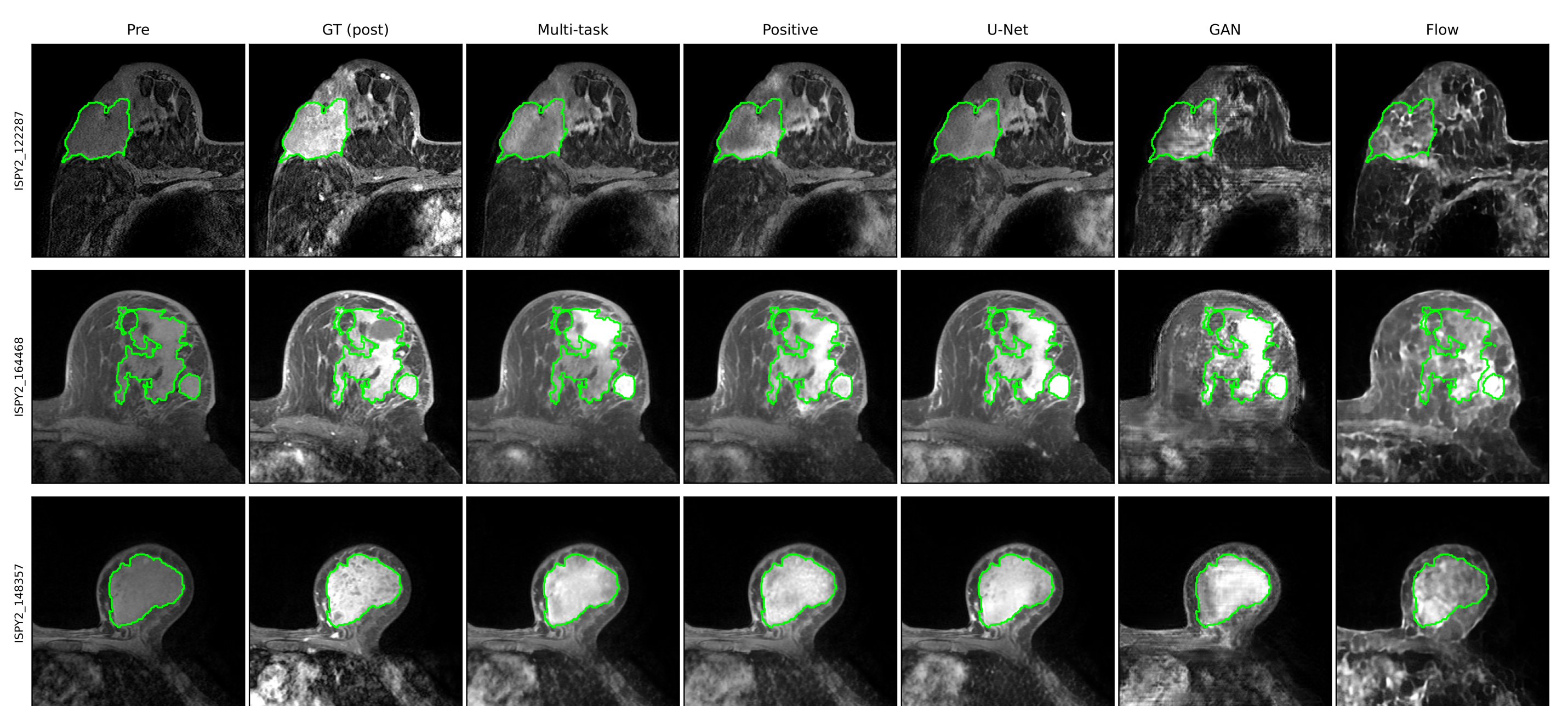


Fig. 2 Largest-tumour cases: pre-contrast, ground truth and four models (mask in lime). Regression reproduces the enhancement faithfully if smoothly; GAN and flow invent texture and lose the lesion.

Limitations: one conventionally parameterised configuration per generative family (no residual variant; the GAN has no skip connections); Figs. 3-4 are single runs; DICE and AUROC inherit the frozen challenge models' errors.